

Ayush Soni

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(Open to Relocate)

Data Scientist | Data Analytics Specialist with 6+ years of experience delivering **data solutions** in technology and financial services. Proven expertise of designing **machine learning models**, deploying **NLP/LLM** solutions, and building scalable data pipelines to optimize decision-making. Skilled in **Python, SQL, Snowflake, AWS, Tableau, and Power BI** with hands-on experience in **Data Engineering and AI**. Strong background in financial analytics, risk modeling, and predictive forecasting with projects recognized at **UBS and Accenture**. Seeking to apply advanced AI and data engineering expertise to drive innovation at leading organizations.

CERTIFICATIONS AND AWARDS

- **Meta Data Analyst**, Issued by Meta - [Link](#)
- **Google Advanced Data Analytics**, Issued by Google - [Link](#)
- **Data Visualization in Tableau & Power BI**, Udemy - [Link](#)
- **Beijing Air Quality Index Forecasting**: Selected as one of the top final projects and hosted on the course website for the Machine Learning course, showcasing exemplary work and recognition by Professor. - [Link](#)
- Received **Q2, FY20 Accenture Team Award in the Client & Customer category** for consistently delivering value to Clients, People, and Organizations by bringing down technical issues by 84%. - [Link](#)

PROFESSIONAL EXPERIENCE

DISH NETWORK (BOOST MOBILE)

Senior Data Scientist

DENVER, CO

Sep 2024 - Present

Delivered analytics and ML solutions that modernized sales forecasting, automated executive insights with Snowflake Cortex AI, and optimized retail operations across 3,000+ locations.

- Built and integrated **Generative AI agents** using **Databricks-hosted LLMs** that enable executives to **query KPIs in natural language**, retrieve live insights, and summarize performance reports — **reducing analytics turnaround time by 70 %**.
- Engineered executive-facing analytics platform integrating **Tableau and Snowflake**, applying LOD calculations and in-stock prediction metrics to monitor device sales — accelerated decision-making speed by **30%** and improved leadership adoption of insights
- Automated end-to-end **Snowflake** pipelines for **3,000+** retail locations, enabling near real-time stock availability monitoring and reducing manual reporting effort by **40%**
- Designed and deployed **ML models** for sales forecasting, price elasticity estimation, and campaign ROI analysis, improving forecast accuracy by **10–15%** and driving multi-million-dollar pricing strategy optimization
- Partnered with cross-functional teams (marketing, supply chain, retail ops) to reduce stockouts and optimize distribution to ensure **100% accuracy** in executive KPIs and reports by resolving critical data integrity issues

THE PARTNERSHIP CENTER LTD.

Data Scientist

Cincinnati, OH

Jun 2024 - Sep 2024

Accelerated investment performance analysis by deploying machine learning models, migrating enterprise data to Snowflake, and automating ETL workflows for scalable, production-ready insights.

- Developed and deployed a **Python-based regression model** to optimize resource utilization, reducing forecast error by 20% and improving schedule tracking efficiency by 40%.
- Automated **ETL workflows using Alteryx and Snowflake**, enabling repeatable, production-ready pipelines and delivering faster insights for portfolio performance monitoring.
- Built interactive **Power BI** dashboards with advanced **DAX calculations** and custom visuals to analyze investment portfolios, highlighting key performance trends for portfolio managers.
- Collaborated with cross-functional teams to integrate **ML** outputs into business processes, ensuring adoption of predictive insights for resource planning and operational efficiency.

UBS

Data Scientist (Quantitative Research & NLP/AI)

Chicago, IL

Jan 2023 - May 2024

Advanced quantitative research and risk modelling by applying AI/ML forecasting, NLP (FinBERT), and automation to financial datasets, enabling more accurate predictions and operational efficiency in investment decision-making.

- Revamped Regression models (**XGBoost**) with **24 months'** rolling data, achieving an impressive **62.40% hit rate** on weekly bond market predictions and strengthening investment decision support
- Automated portfolio data collection from **YODA and Bloomberg**, reducing manual effort by **70%** and improving governance model validation efficiency for US multi-sector teams through **Python** Script automation
- Boosted the accuracy of forecasting models for global mutual funds by **33%** through advanced **data validation and feature engineering** on **Bloomberg** Datasets
- Built and deployed an **NLP sentiment analysis prototype** using **FinBERT**, achieving **87%** accuracy in classifying financial news and integrating insights into credit risk assessments.
- Automated financial reporting tasks on a credit **rating platform** by leveraging **AI-driven analytics** to monitor industry risk trends, conduct competitive benchmarking, and streamline business reporting—achieving a **75%** increase in operational efficiency, reduced manual effort, and improved accuracy of credit risk strategies in investment banking
- Designed predictive pipelines in **Python (NumPy, Pandas, Scikit-Learn, TensorFlow, Keras)** and **SQL** to model stock market data, enabling quantitative analysts to forecast equity performance and inform trading strategies.
- Delivered executive-facing visual analytics in **Tableau** to present risk exposure, forecasting accuracy, and portfolio performance to senior stakeholders

ACCENTURE

Machine Learning Associate

Client: OP Financial, Finland

Pune, India

Jun 2018 - Jul 2021

Empowered OP Bank's operations by leveraging data-driven insights to optimize card limits, improve data accuracy, and increase revenue by 23%, alongside achieving a 40% efficiency increase

- Designed dynamic credit/debit card limit management models by integrating customer account activity, loan history, and credit scores, increasing revenue by **23%** and operational efficiency by **40%**.
- Applied **K-means clustering** and rule-based models to segment customers by risk-adjusted revenue potential, enabling targeted marketing and personalized engagement strategies.
- Conducted **A/B testing** on marketing campaigns, using ML-driven customer segments to improve satisfaction and response rates.
- Performed statistical analysis (**t-tests, chi-square, ANOVA, correlation, regression**) to uncover patterns and improve decision-making in credit and debit card portfolios.
- Built Tableau dashboards and designed **SQL** views to track key financial KPIs, improving market data accuracy by **25%**.
- Optimized **ETL workflows** and SQL queries in MySQL and SQL Server to streamline reporting pipelines, driving an **8%** increase in profitability.

Client: Allianz Bank, Germany

Delivered data-driven insights empowering cross-functional teams to enhance operational efficiency, optimize resource allocation, and drive strategic decision-making for improved client service delivery

- Enhanced client service delivery and operational efficiency by combining machine learning models, cloud migration, and advanced reporting solutions to optimize resource allocation and business decision-making
- Developed an **XGBoost classifier** for customer churn prediction (**F1 score: 0.84**), identifying key loyalty drivers and informing retention strategies that boosted customer engagement and revenue
- Built a **Python regression model** to improve resource utilization forecasts, reducing forecast error by 20% and increasing scheduling efficiency by **40%**
- Migrated enterprise data from **DB2/Teradata to AWS Cloud**, ensuring scalability, cost-effectiveness, and integrity of analytics infrastructure
- Designed executive dashboards in **Tableau and Power BI** with advanced **DAX** calculations, improving operational insights and reducing manual reporting time by **10%**
- Standardized experimentation frameworks to minimize bias and improve model interpretability, increasing accuracy of analytical outcomes by **90%**

PROJECTS

Hands-on AI/ML projects showcasing applied expertise in anomaly detection, NLP, recommender systems, and time-series forecasting

Anomaly Detection in Credit Card Transactions

- Designed a fraud detection system analyzing **29,000+** credit card transactions using **Python and Power BI**, with automated alerts for anomalies
- Enabled **real-time fraud monitoring**, improving detection accuracy and reducing response times for financial risk teams.

Beijing Air Quality Index Forecasting

- Formulated a machine learning-based air quality prediction strategy utilizing historical data, generating a **24-hour** air quality forecast
- Implemented **Neural Network, Gaussian Naïve Bayes, Linear Regression with Gradient Descent**, and **Multi-class Logistic Regression** Methods, evaluating method accuracies based on RMSE after hyper-parameter tuning

NLP Sentiment Analysis for Financial News (FinBERT Prototype)

- Implemented **FinBERT**, a transformer-based **LLM**, to classify financial news sentiment with **87%** accuracy.
- Integrated sentiment outputs into credit risk assessments, strengthening investment decision-making processes.

Recommendation System using Amazon Reviews

- Engineered **Collaborative Filtering** and **Content-based recommendation** system using Amazon Review data
- Implemented ML algorithms **KNN, SVD**, and **ALS** for recommending products to new and existing users with an accuracy of **89.58%**

SKILLS

- **Programming & Databases:** Python, R, SQL, Base SAS, PySpark, Snowflake, Databricks, Hive
- **Data Engineering & Visualization:** Tableau, Power BI, QlikView, ALteryx, SAS Enterprise, Advanced Excel, ETL/ELT Pipelines
- **AI/ML Algorithms & Python Libraries:** Regression & Classification, Random Forest, XGBoost, Neural Networks, SVM, KNN, Clustering, Deep Learning (PyTorch, TensorFlow, Keras), NLP/LLMs (Hugging Face, FinBERT), Recommender Systems, Forecasting & Time Series Models, Scikit-Learn, NumPy, Pandas, SciPy, Matplotlib, Seaborn
- **MLOps & Deployment:** MLflow, Docker, Kubernetes, Airflow, Jenkins, Git/GitHub, CI/CD Pipelines
- **Cloud Platforms:** AWS (S3, EC2, SageMaker, Redshift, Lambda, Glue), GCP (BigQuery, Dataflow), Azure (Data Factory, SQL Database)
- **Development/Other Tools:** Jira, VS Code, Eclipse, ServiceNow, Bloomberg Terminal

EDUCATION

NORTHEASTERN UNIVERSITY	Boston, USA
<i>Master of Science in Data Analytics Engineering</i>	Sep 2021 - Dec 2023
<i>Relevant Coursework:</i> Data Management, Data Visualization, Statistical Methods, Data Mining, Machine Learning, Deep Learning	
NARSEE MONJEE INSTITUTE OF MANAGEMENT STUDIES	Mumbai, India
<i>Bachelor of Technology in Computer Engineering</i>	Jul 2014 - Aug 2018